

Ava Robillard

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EDUCATION

Master of Environmental Data Science, 4.00 GPA

University of California, Santa Barbara- Bren School of Environmental Science & Management

Related Courses: Geospatial Analysis, Machine Learning, Data Visualization and Communication, Databases and Data Management, Environmental Policy Evaluation, Modeling Environmental Systems

Bachelor of Science in Plant Sciences, Minor in Data Science, 3.95 GPA (May 2025)

Cornell University, Ithaca, NY

Related Courses: Data Analysis in Ecology, Remote Sensing, Introduction to Computing: Design and Development, Linear Algebra, Probability Models, Urban Plants and Public Health, Forest Ecology

Leadership: Plant Science Peer Mentor, Cornell Botanical Gardens Horticultural Enterprises

Honors: Pi Alpha Xi National Honor Society for Horticulture

DATA SCIENCE PROJECT EXPERIENCE

Rincon's Bio Weaver Tool- From Disparate Data to Actionable Analysis (1/26-6/26)

Client: Rincon Consultants, Inc. | Master's Capstone Project

- Developed a data pipeline integrating multiple biological databases to connect backend species occurrence data retrieval to a user-facing interface, reducing manual processing time
- Designed an interactive web application that accepts user-defined project parameters and generates species occurrence maps, visualizations, and structured tables for regulatory reporting
- Collaborated with a 3-person capstone team via GitHub to implement version-controlled workflows with comprehensive documentation to enable independent client use

NYC Street Tree Spatial Analysis | Data Visualization & Communication Course Project (1/26-3/26)

- Built a data-driven infographic examining urban tree equity across NYC boroughs, joining NYC Open Data's 600,000-record Tree Census with American Community Survey and Census TIGER/Line spatial data
- Published reproducible code on GitHub designed for future integration with the 2025 NYC Tree Census
- Infographic selected for permanent display at the Bren School of Environmental Science & Management

PROFESSIONAL EXPERIENCE

Computational Biology REU Intern – National Science Foundation, Chattanooga, TN (5/24-7/24)

- Processed and analyzed Landsat satellite imagery in Google Earth Engine to develop an Urban Heat Island Index using normalized indices for vegetation greenness and land surface temperature
- Designed spatial data maps in ArcGIS Pro, revealing temporal trends to assist in informing urban planning strategies for climate resilience
- Presented research at the ASPRS International Technical Symposium (2024): Spatio-temporal Analysis of Urban Heat Island Effect Using Remote Sensing, GIS, and Google Earth Engine

Research Assistant II – Evolution of Plant Form & Function Lab, Cornell University, Ithaca, NY (2/22-5/25)

- Conducted bioinformatics analyses, including plastome annotation and alignments, single nucleotide polymorphism (SNP) calling, and phylogenetic inference using Bash scripts
- Extracted and quantified DNA using gel electrophoresis and Qubit for perennial species of *Glycine*
- Organized seed stocks and germinated seedlings for DNA sequencing of genetic diversity
- Co-author on Botany 2023 presentation: B Landis, A Robillard, A Mahan, and JJ Doyle. Overcoming phylogenomic challenges of nested polyploidy events in *Glycine* using genome-wide data.

SKILLS

Languages & Libraries: Python (NumPy, pandas, GeoPandas, xarray, rioarray, matplotlib), R (tidyverse, sf, terra, tmap, ggplot2), SQL, Bash

Tools & Platforms: Git/GitHub, VSCode, Jupyter, Quarto, ArcGIS, Google Earth Engine, Microsoft Office Suite, Google Suite, Affinity